

# Samuel Philip Ovalle Rodriguez

Desarrollador Full-Stack · JavaScript · React.js · Node.js · PHP · REST API

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Platzi](#)

## PROFESSIONAL PROFILE

Full-Stack Developer with over 4 years of experience designing and implementing end-to-end web solutions: from SPA application architecture with React.js and Vue.js to REST APIs, business logic in PHP/Laravel, and persistence in MySQL. Experience in international environments (Colombia, USA) and in full-cycle projects with direct client management. Proven ability to solve complex engineering problems: implementation of search algorithms (A\*), performance optimization, and object-oriented architecture in my own projects. Committed to clean code, best practices, and continuous learning. Native Spanish speaker; English B2 level.

## TECHNICAL SKILLS

### Frontend

|                   |              |
|-------------------|--------------|
| HTML5 / CSS3      | Advanced     |
| JavaScript (ES6+) | Advanced     |
| TypeScript        | Advanced     |
| React.js          | Advanced     |
| Vue.js            | Advanced     |
| Responsive Design | Advanced     |
| Tailwind CSS      | Intermediate |
| SPA / REST API    | Advanced     |
| WordPress         | Advanced     |

### Backend

|                       |              |
|-----------------------|--------------|
| PHP                   | Advanced     |
| Laravel (MVC)         | Intermediate |
| Node.js               | Intermediate |
| API REST / JWT        | Intermediate |
| Authentication / Auth | Intermediate |
| Python                | Intermediate |

### Databases & DevOps

|                    |              |
|--------------------|--------------|
| MySQL / SQLite     | Intermediate |
| Git / GitHub       | Advanced     |
| Linux              | Basic        |
| Deployment / CI-CD | Basic        |

### Architecture & Engineering

|                      |              |
|----------------------|--------------|
| OOP / Class Design   | Advanced     |
| Algorithms (A*, BFS) | Advanced     |
| Memory management    | Intermediate |
| MVC Patterns         | Intermediate |
| UX/UI Design         | Advanced     |
| Arduino / Embedded   | Intermediate |

### Tools

VS Code · Figma · Adobe XD · Illustrator · Photoshop · Blender

## PROFESSIONAL EXPERIENCE

### Full-Stack Developer

Skygreen · Florida, USA (Remote)

Jan 2025 – Apr 2025

- I designed and developed the complete corporate web platform—from architecture to deployment—implementing a responsive frontend, business logic in PHP, and data persistence in MySQL.
- I designed the relational data model and optimized SQL queries to ensure integrity and performance in production.
- I built and integrated a REST API that connects the SPA frontend with the backend, applying separation of responsibilities and MVC patterns.
- I ran complete QA and debugging cycles, reducing the error density before delivery to the client.

- Stack: HTML5, CSS3, JavaScript, PHP, MySQL, Git.

Ref: Jeisson Roger Ovalle Aguilar (CEO) · +1 (786) 744-0419

## Freelance Full-Stack Developer

Refiérello SAS · Bogotá, Colombia

Jun 2023 – Dic 2024

- I translated client designs into functional web applications, implementing responsive frontends and integrating PHP backends with MySQL persistence, reducing delivery times through a structured development workflow.
- I developed data management modules with server-side validation and input sanitization to prevent SQL injections.
- I maintained direct communication and feedback loops with the client throughout the project lifecycle, ensuring alignment with business requirements.
- Stack: HTML5, CSS3, JavaScript, PHP, MySQL, Git.

Ref: Miguel Angel Rodríguez (CEO) · +57 316 741 2647

## FEATURED TECHNICAL PROJECTS

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### Thinkin Ships — AI Game Engine

2025

JavaScript · HTML5 Canvas · OOP · Personal Project

- I developed my own object-oriented JavaScript game engine with a decoupled class architecture (ships, board, AI, renderer), FPS control, and Canvas rendering without external dependencies.
- I implemented the A\* search algorithm for AI autonomous navigation, optimized with Manhattan heuristics, redundant path pruning, and open/closed list structures to minimize memory usage in long sessions.
- I designed a path compression system using 'runs' and manual object lifecycle management to prevent memory leaks, achieving stable performance throughout the game.
- 100% original code and art; extensible architecture that allows adding new ships, maps, and rules without modifying the core of the engine.

### Movi Grip — Interactive robot with vision and remote control

2024

Python · Arduino · Blender · Electronics · Personal Project

- I developed the Bluetooth communication layer between Python (host) and an Arduino microcontroller for real-time command transmission with minimal latency.
- I programmed the Arduino firmware to interpret commands and control movement, an articulated arm with servomotors, and an integrated screen with multiple visual patterns.
- I modeled and printed the chassis in Blender, integrating real physical constraints of the electronics, combining software engineering, electronics, and 3D design into a single pipeline.

## EDUCATION

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### Software Analysis and Development Technologist

2025 – Ongoing

SENA · Bogotá, Colombia

### Continuing education in software developmentPlatzi

[View certifications → platzi.com/p/sfovallerodriguez](https://platzi.com/p/sfovallerodriguez)

## LANGUAGES

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**Spanish:** Native

**English:** Intermediate — B2 (fluent technical reading)